

θ -TaN Literature, Simulation, and MLIP Review

Research-delivery edition - 15 August 2026. This review focuses on the hexagonal theta phase of tantalum nitride (θ -TaN), with priority given to direct material-property and atomistic-simulation papers.

Executive Summary

θ -TaN is a metallic, topological semimetal candidate whose unusually high heat transport is central to the literature. The established theoretical transport line spans acoustic-optical phonon-gap arguments, first-principles Boltzmann transport including higher-order scattering, and pressure-dependent phonon-electron scattering. Recent experimental single-crystal reports make the transport result substantially more concrete.

For MD-model development, two θ -TaN-specific machine-learned interatomic-potential studies were found: a 2025 Deep Potential (DP) transport model and a 2026 Neuroevolution Potential (NEP) deformation/fracture model. These are useful baselines but have different intended domains; neither alone is a demonstrated general-purpose potential across defects, free surfaces, thermal transport, and extreme pressure.

Search Method and Scope

Firecrawl Research semantic paper searches were run for θ -TaN material properties, first-principles methods, phonons/thermal transport, pressure, molecular dynamics, Deep Potential, and NEP. Candidate records were expanded and metadata or in-body passages were checked for the two MLIP papers. The bibliography below separates direct θ -TaN records from broader TaN context. It is a broad web-indexed search, not a claim of an exhaustive database census.

Direct θ -TaN Paper Set

The following records are direct matches and form the recommended reading order: topology/structure, transport theory, pressure response, experiments, then MLIPs.

Direct θ -TaN Properties and Transport

- [1] H. Weng, C. Fang, Z. Fang, and X. Dai, *Topological Semimetals with Triply Degenerate Nodal Points in θ -phase Tantalum Nitride* (2016). arXiv:1604.08467. <https://arxiv.org/abs/1604.08467>. Establishes the triply-degenerate-nodal-point/topological-semimetal framing for θ -TaN.
- [2] S.-D. Guo and B.-G. Liu, *Ultra-high lattice thermal conductivity in topological semimetal TaN caused by a large acoustic-optical gap* (2018). DOI:10.1088/1361-648X/aaab32. First-principles transport study connecting the phonon spectrum to high lattice thermal conductivity.
- [3] A. Kundu et al., *Ultra-high Thermal Conductivity of θ -Phase Tantalum Nitride*, Physical Review Letters 126, 115901 (2021). DOI:10.1103/PhysRevLett.126.115901. First-principles/BTE treatment that includes three- and four-phonon, electron-phonon, and isotope scattering.
- [4] A. Kundu et al., *Electron-Induced Nonmonotonic Pressure Dependence of the Lattice Thermal Conductivity of θ -TaN*, Physical Review Letters 132, 116301 (2024). DOI:10.1103/PhysRevLett.132.116301. Predicts κ_{ph} increasing toward about 60 GPa before decreasing as electronic states increase phonon-electron scattering.
- [5] S. Li et al., *Metallic θ -phase tantalum nitride has a thermal conductivity triple that of copper*, Science (2026). DOI:10.1126/science.aeb1142. Experimental realization of single-crystalline θ -TaN and high thermal conductivity.
- [6] Y. Liu et al., *High thermal conductivity in metallic θ -TaN single crystals*, National Science Review (2026). DOI:10.1093/nsr/nwag106. Complementary single-crystal experimental report.

Structure, Mechanics, and Pressure

- [7] J. Chang et al., *Structure and mechanical properties of tantalum mononitride under high pressure: A first-principles study*, AIP Advances (2012). DOI:10.1063/1.4759279. Includes θ -phase behavior among stable TaN phases and is useful for pressure/elastic training configurations.

For a training project, papers [2]-[4] define the phonon and transport validation targets, while [7] is a useful initial reference for volume/pressure paths. Do not transfer conclusions for other TaN polymorphs to θ -TaN without phase-specific checks.

Existing θ -TaN MLIPs

Deep Potential for Thermal Transport

[8] Z. Zong, Y. Qin, J. Zhan, H. Fang, and N. Yang, *Machine learning potential for predicting thermal conductivity of θ -phase and amorphous Tantalum Nitride* (2025). arXiv:2508.03297. <https://arxiv.org/abs/2508.03297>.

This paper trains θ -TaN and amorphous-TaN Deep Potential models with active learning in DP-GEN. VASP reference data use a 600 eV plane-wave cutoff; exploration uses ensembles of four independently seeded DP models and force-model deviation for candidate selection. The θ -TaN energy RMSE reported in the retrieved passages is 0.505 meV/atom, and the force RMSE is 0.026 eV/Å. The work validates θ -TaN phonon dispersion and volume scaling against DFT, then uses EMD and NEMD in LAMMPS for bulk and nanofilm thermal transport.

NEP for Tensile Response and Fracture

[9] C. Cao, H. Li, and S. Cao, *Anisotropic Tensile Strength and Fracture Mechanism of θ -TaN: A Machine-Learning Potential Molecular Dynamics Study* (2026). arXiv:2607.27608. <https://arxiv.org/abs/2607.27608>.

This paper trains a NEP in GPUMD using VASP PBE/PAW references (600 eV cutoff) and active learning. The final training data contain 1,053 configurations and 115,752 atoms: 300-900 K thermal snapshots, $\pm 6\%$ lattice perturbations, shear, uniaxial 1-25% tensile paths along a and c, and generalized stacking faults. A separate 400-configuration hydrostatic-compression test set spans 1, 2, 5, and 10 GPa at 300-900 K. The reported training RMSEs are 0.9 meV/atom for energy and 34.5 meV/Å for forces.

Practical Status

Yes: θ -TaN-specific MLIPs already exist. The DP paper is strongest as a transport-oriented baseline; the NEP paper is strongest for high-strain and fracture mechanics. The retrieved sources did not establish an openly archived, ready-to-download parameter file or a complete raw DFT dataset for either model. Treat availability as something to confirm with the authors or linked supplementary material before planning reuse.

Recommended MLIP Training Plan

Use one uniform DFT protocol rather than merging published labels. A defensible starting baseline is PBE/PAW, 600 eV, explicit k-point convergence, and consistent electronic-smearing settings. Record code version, pseudopotentials, convergence tests, and every failed or out-of-domain configuration.

Configuration coverage

Include equilibrium EOS and strained bulk cells; finite-temperature NVT/NPT snapshots; hydrostatic compression and expansion; uniaxial and shear paths; generalized stacking faults; vacancies/interstitials/antisites; surfaces and nanofilms if relevant; and candidate competing phases only if the model is deliberately multi-phase. Include acoustic phonon behavior and thermal-expansion checks if κ is an objective.

Validation split

Hold out complete trajectory families, not random frames: pressure trajectories, large-strain paths, defect cells, and temperatures. Report energy, force, and virial/stress errors separately. Benchmark EOS, elastic constants, phonons, stacking-fault energies, and stress-strain curves against direct DFT. For transport, test size, time, and sampling convergence before comparing EMD/NEMD to BTE or experiment.

Adjacent/Contextual References

S.-D. Guo et al., *First principles investigation on the ultra-incompressible and hard TaN* (2009), DOI:10.1002/jcc.21234. Broad TaN polymorph context; not θ -TaN-specific.

Z. Zong et al. [8] also covers amorphous TaN and is useful if a crystalline-amorphous interfacial model is required.

Source links use DOI resolvers or arXiv records so they are usable independently of the Firecrawl search session.